

Exam – Reference Solution

Numerical Programming I

February 6, 2026

Name: _____

Question:	1	2	3	4	Total
Points:	5	5	9	0	19
Score:					

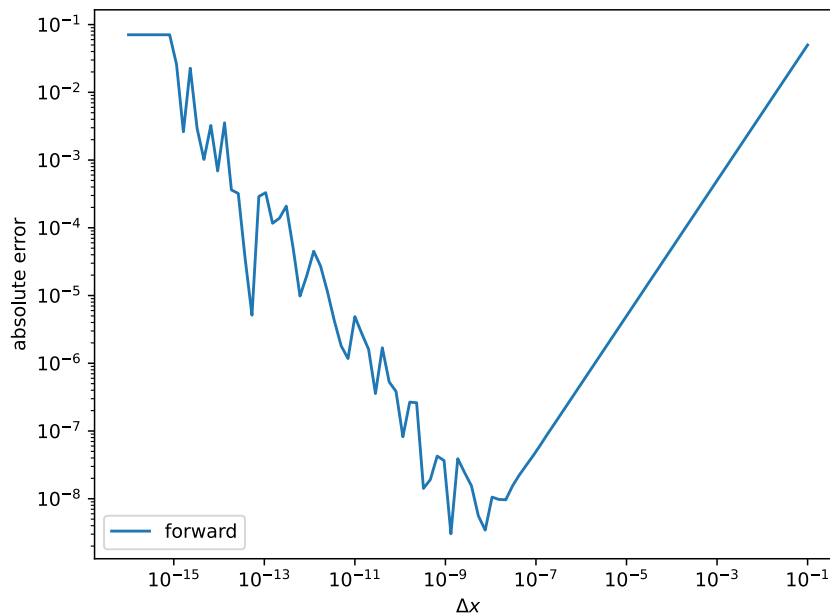
Mixed Short Questions

1. (a) What does $(1.0+10^{-17})-1.0$ evaluate to in double-precision floating point arithmetic (with machine precision $\epsilon_{\text{mach}} \sim 10^{-16}$)?
- ☐ 10^{-17} is correctly calculated.
 - ☐ 10^{-17} cannot be represented in double-precision, the expression results in undefined behavior.
 - ☒ **The expression evaluates to zero, the mantissa of 1.0 does not have enough digits to capture 10^{-17} .**

Solution

Since $10^{-17} < \epsilon_{\text{mach}}/2$ relative to 1.0, the addition $1.0 + 10^{-17}$ results in exactly 1.0 due to rounding (machine epsilon is the smallest number such that $1 + \epsilon \neq 1$). Subtracting 1.0 yields 0.0.

2. (b) Consider a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$. We have written a program which approximates $f'(x)$ using the forward difference $\frac{f(x+\Delta x)-f(x)}{\Delta x}$. Below, the absolute error $\left|f'(x) - \frac{f(x+\Delta x)-f(x)}{\Delta x}\right|$ over Δx is plotted. Mark the region where the floating point error is dominant and the region where the truncation error is dominant.



Solution

Left Region (jagged/noisy lines): Floating point cancellation error is dominant (as $\Delta x \rightarrow 0$).

Right Region (straight slope 1 line): Truncation error is dominant (mathematical approximation error).

- 2 (c) Does gradient descent always converge to the **global** minimum? Answer with yes or no and give a short justification.

Solution

No. Gradient descent follows the steepest descent direction locally. It will converge to a **local** minimum depending on the initialization. It only guarantees global convergence for strictly convex functions.

Second order derivative approximation

2. For a twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ we would like to find a finite difference approximation for the second derivative $f''(x)$.

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- (a) Approximate $f(x + \Delta x)$ and $f(x - \Delta x)$ by Taylor expansion up to (including) third order around x .

Solution

The Taylor expansions are:

$$\begin{aligned} f(x + \Delta x) &= f(x) + f'(x)\Delta x + \frac{1}{2}f''(x)\Delta x^2 + \frac{1}{6}f'''(x)\Delta x^3 + \mathcal{O}(\Delta x^4) \\ f(x - \Delta x) &= f(x) - f'(x)\Delta x + \frac{1}{2}f''(x)\Delta x^2 - \frac{1}{6}f'''(x)\Delta x^3 + \mathcal{O}(\Delta x^4) \end{aligned}$$

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- (b) Using these approximations find an approximation for $f''(x)$ based on $f(x)$, $f(x + \Delta x)$ and $f(x - \Delta x)$. What is the order of the error term? *Hint*: You need to combine both Taylor expansions such that the first and third order derivatives cancel.

Solution

Summing the two equations from part (a):

$$f(x + \Delta x) + f(x - \Delta x) = 2f(x) + f''(x)\Delta x^2 + \mathcal{O}(\Delta x^4)$$

(Note that the odd terms f' and f''' cancel out).

Solving for $f''(x)$:

$$\begin{aligned} f''(x)\Delta x^2 &= f(x + \Delta x) - 2f(x) + f(x - \Delta x) + \mathcal{O}(\Delta x^4) \\ f''(x) &= \frac{f(x + \Delta x) - 2f(x) + f(x - \Delta x)}{\Delta x^2} + \mathcal{O}(\Delta x^2) \end{aligned}$$

The order of the error term is $\mathcal{O}(\Delta x^2)$ (second order).

1

- (c) For $f(x) = x^2$ and $\Delta x = 0.1$ approximate $f''(0)$ and compare your approximation to the analytical value.

Solution

Analytical: $f(x) = x^2 \implies f'(x) = 2x \implies f''(x) = 2$. Thus $f''(0) = 2$.

Approximation: Using the formula derived above at $x = 0$:

$$\begin{aligned} f''(0) &\approx \frac{f(0.1) - 2f(0) + f(-0.1)}{(0.1)^2} \\ &= \frac{0.01 - 0 + 0.01}{0.01} = \frac{0.02}{0.01} = 2 \end{aligned}$$

The approximation matches the analytical value exactly (because the error term depends on $f^{(4)}$, which is 0 for a quadratic).

Convergence of Newton's root finding method

3. In the lecture we have stated:

If g is continuously differentiable in a neighborhood around a root x^* with nonzero derivative at x^* , then there exists a neighborhood around x^* where Newton's iteration converges to x^* for all starting values.

Now consider the function

$$g(x) = x(1 + x^2)^{-\frac{1}{2}} \quad (1)$$

- [1] (a) What is the root x^* of g where $g(x^*) = 0$?

Solution

$g(x) = \frac{x}{\sqrt{1+x^2}}$. Since the denominator $\sqrt{1+x^2} \geq 1$ never vanishes, the only solution is the numerator being zero.

$$x^* = 0$$

- [2] (b) Show that the derivative of g is $g'(x) = (1 + x^2)^{-\frac{3}{2}}$. *Hint:* Calculate the derivative by product rule and factor out $(1 + x^2)^{-\frac{3}{2}}$.

Solution

Using the product rule on $x \cdot (1 + x^2)^{-1/2}$:

$$\begin{aligned} g'(x) &= 1 \cdot (1 + x^2)^{-\frac{1}{2}} + x \cdot \left(-\frac{1}{2}\right) (1 + x^2)^{-\frac{3}{2}} \cdot (2x) \\ &= (1 + x^2)^{-\frac{1}{2}} - x^2(1 + x^2)^{-\frac{3}{2}} \end{aligned}$$

Factor out $(1 + x^2)^{-\frac{3}{2}}$:

$$\begin{aligned} &= (1 + x^2)^{-\frac{3}{2}} [(1 + x^2)^1 - x^2] \\ &= (1 + x^2)^{-\frac{3}{2}} \cdot 1 = (1 + x^2)^{-\frac{3}{2}} \end{aligned}$$

- [1] (c) Calculate $g'(x^*)$. Is there a neighborhood around x^* where Newton's iteration converges to x^* for all starting values inside that neighborhood?

Solution

$$g'(0) = (1 + 0^2)^{-\frac{3}{2}} = 1$$

Since $g'(x^*) = 1 \neq 0$, the condition of the theorem holds. **Yes**, there exists such a neighborhood.

- 2 (d) Show that the Newton step

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)} \quad (2)$$

is given by $x_{k+1} = -x_k^3$.

Solution

$$\begin{aligned} \frac{g(x)}{g'(x)} &= \frac{x(1+x^2)^{-\frac{1}{2}}}{(1+x^2)^{-\frac{3}{2}}} \\ &= x(1+x^2)^{-\frac{1}{2}}(1+x^2)^{\frac{3}{2}} \\ &= x(1+x^2)^1 = x + x^3 \end{aligned}$$

Therefore:

$$x_{k+1} = x_k - (x_k + x_k^3) = -x_k^3$$

- 3 (e) Consider the following cases for the initialization of the Newton iteration

- $x_0 = 0.5$
- $x_0 = -1.0$
- $x_0 = 2.0$

and analyze if the iteration converges, diverges or oscillates.

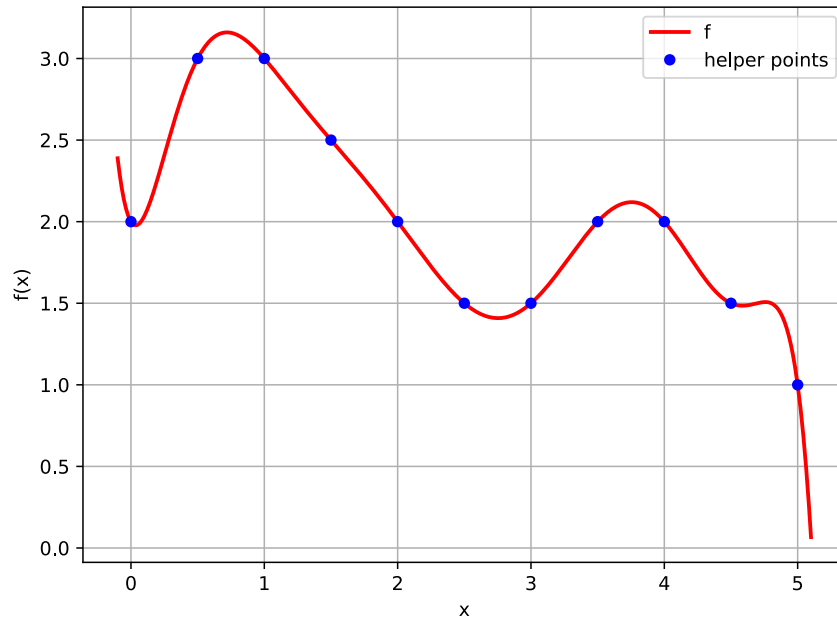
Solution

Using $x_{k+1} = -x_k^3$:

- $x_0 = 0.5$: $x_1 = -0.125$, $x_2 = -(-0.125)^3 \approx 0.002$. Values get smaller rapidly. **Converges** to 0.
- $x_0 = -1.0$: $x_1 = -(-1)^3 = 1.0$, $x_2 = -(1)^3 = -1.0$. **Oscillates** between -1 and 1.
- $x_0 = 2.0$: $x_1 = -(2)^3 = -8$, $x_2 = -(-8)^3 = 512$. Values grow rapidly. **Diverges**.

(*Bonus*) Numerical integration by hand

4. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ plotted below.



We want to approximate the integral

$$\int_0^5 f(x) dx \quad (3)$$

Here, assume the subdivision

$$\int_0^5 f(x) dx = \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} f(x) dx \quad (4)$$

with $N = 5$ and $x_0 = 0.0, x_1 = 1.0, x_2 = 2.0, x_3 = 3.0, x_4 = 4.0, x_5 = 5.0$.

And consider the following *quadrature rules*:

- midpoint rule:

$$\int_{x_i}^{x_{i+1}} f(x) dx = \underbrace{(x_{i+1} - x_i) f\left(\frac{x_i + x_{i+1}}{2}\right)}_{=: M_i} + \mathcal{O}((x_{i+1} - x_i)^3) \quad (5)$$

- trapezoidal rule:

$$\int_{x_i}^{x_{i+1}} f(x) dx = \underbrace{(x_{i+1} - x_i) \left(\frac{f(x_i) + f(x_{i+1})}{2}\right)}_{=: T_i} + \mathcal{O}((x_{i+1} - x_i)^3) \quad (6)$$

2 (bonus)

- (a) Approximate $\int_0^5 f(x) dx$ using the midpoint rule as given above.

Solution

For the Midpoint rule with $N = 5$, we need values at $x = 0.5, 1.5, 2.5, 3.5, 4.5$.
Reading the blue dots on the graph:

- $f(0.5) = 3.0$
- $f(1.5) = 2.5$
- $f(2.5) = 1.5$
- $f(3.5) = 2.0$
- $f(4.5) = 1.5$

The midpoint approximation yields

$$M = \Delta x \sum f(\text{midpoints}) = 1.0 \cdot (3.0 + 2.5 + 1.5 + 2.0 + 1.5) = \mathbf{10.5} \quad (7)$$

2 (bonus)

- (b) Approximate $\int_0^5 f(x) dx$ using the trapezoidal rule as given above.

Solution

For the Trapezoidal rule, we need values at integers $x = 0, 1, 2, 3, 4, 5$. Reading the graph:

- $f(0) = 2.0$
- $f(1) = 3.0$
- $f(2) = 2.0$
- $f(3) = 1.5$
- $f(4) = 2.0$
- $f(5) = 1.0$

The trapezoidal approximation yields

$$\begin{aligned} T &= \Delta x \sum_{i=0}^4 \frac{f(x_i) + f(x_{i+1})}{2} \\ &= \Delta x \left[\frac{f(0) + f(5)}{2} + \sum_{i=1}^4 f(x_i) \right] \\ &= 1.0 \cdot \left[\frac{2.0 + 1.0}{2} + (3.0 + 2.0 + 1.5 + 2.0) \right] \\ &= \mathbf{10.0} \end{aligned} \tag{8}$$